

Practical Assignment 2

Sr. No	Practicals	Type
1	Create three variables: first_name (assign a text string), age (assign an integer), and is_student (assign a boolean). Print all three variables on separate lines.	Easy
2	Assign the value 15 to variable a and 4 to variable b. Write expressions to find and print their sum, product, and the result of a raised to the power of b.	Easy
3	Create a variable total_days = 45. Use the modulus operator (%) to find how many days are left over after converting these days into full weeks (7 days per week).	Easy
4	Take two integer inputs from the user. Write a program that uses relational operators to check if the first number is greater than or equal to the second number. Print True or False.	Easy
5	Create two boolean variables: has_compiled = True and passed_tests = False. Use the logical and operator to print whether the code is fully ready for deployment.	Easy
6	Create a variable named wallet and initialize it with 100. Using only compound assignment operators (like +=, -=, *=), simulate adding 50, subtracting 20, and then doubling the total amount. Print the final value of wallet.	Moderate
7	Write a program that divides 27 by 5. Print the exact decimal result using normal division, and the whole number result using floor division (//) on the next line.	Moderate
8	Create an integer variable score = 85. Write a single line of code that checks if score is greater than 70 and less than or equal to 90 using relational operators.	Moderate
9	Create a boolean variable is_raining = False and another has_umbrella = True. Write a logical expression that evaluates to True if it is not raining or if the person has an umbrella.	Moderate

10	Ask the user to input two numbers, but accept them as strings. Convert (cast) them into floats inside your program, add them together, and print the final sum.	Moderate
11	A user wants to withdraw a specific amount of money from an ATM (e.g., rupees 235). Because the machine only handles 100 rupees bills, it can only dispense amounts that are perfect multiples of 10. Write a program using arithmetic and relational operators that outputs True if the amount is valid for withdrawal and False if it isn't.	Thinking-Based
12	A bakery recipe requires exactly 3 eggs to make one batch of cookies. The baker looks in the fridge and finds they have total_eggs = 14. Write a program using arithmetic operators to calculate exactly how many full batches they can make, and how many individual eggs will be left over.	Thinking-Based
13	You are programming a smart home display. The outside temperature sensor reads temperature = 18 degrees Celsius. Write an expression combining relational and logical operators to check if the weather is in the "ideal jacket range" (between 15 and 22 degrees inclusive), printing True if it is.	Thinking-Based
14	To enter the VIP section of a festival, a guest must either have a vip_pass (True/False) OR be over the age of 21. However, if they are on the banned_list (True/False), they cannot enter under any circumstances. Create these three variables with sample values and write a single logical expression to determine if access is granted.	Thinking-Based
15	A website registration page requires all new usernames to be secure. The rule states a username must be longer than 5 characters but no longer than 15 characters. Take a string input for a username, calculate its length, and use operators to output whether the username fits the required security rules.	Thinking-Based